

F-65 Megawing

Graphics White Paper v2.1

Document Type: Technical Architecture White Paper

Primary Genre: Combat flight simulation (civil & light combat presentation)

Audience: Systems architects, graphics/engine implementers, C developers, and 45GS02 optimization engineers

Platform: MEGA65 — 45GS02 + VIC-IV

Version: 2.1

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Status: Architecture-review candidate under F65-GAD-001

Controlling parents:

Role	Document
Architecture	F-65 Megawing Revision 1.5.1 — Architecture Invariants
Engine design	F-65 Engine Runtime and Toolchain Design Supplement Draft 0.2
Graphics architecture decision	F65-GAD-001 — Hybrid Affine / Sparse-Polygon World Renderer v0.1
Supporting reference only	MEGA65 Flight Simulation 3D Graphics White Paper Rev 1.0

Nature: Candidate detailed graphics design intended to implement the graphics architecture established by F65-GAD-001. Does not freeze R0-gated numeric limits, display mode, RRB production selection, or production asset formats. Cannot relax Architecture 1.5.1 or Engine 0.2 contracts.

Revision 2.1 focus: Cross-layer occlusion contract; terrain visual/query registration; displayed-world age metrics; atomic view/generation swap; WorldRegistrationRecord atomicity; LOD hysteresis; registration-anchor priority; shedding vs production tiers; mandatory F-65 visual contracts; horizon ownership clarification; **R0-D/E/F downstream scheduling** (R0-A/B/C not retrofitted).

0. Purpose and hierarchy

This white paper defines the **candidate detailed graphics design** intended to implement the graphics architecture established by F65-GAD-001 for F-65 Megawing. It replaces the Mode-7-centered assumptions of Graphics White Paper Rev 1.0 with the hybrid model frozen by F65-GAD-001 and integrates the affine/scanline surface path into the Engine 0.2 sparse polygon pipeline.

- Revision 1.5.1 — Architecture Invariants
 - Engine Runtime and Toolchain Design Supplement 0.2
 - Software 3D pipeline (§8) — sparse geometry basis
- F65-GAD-001 — Hybrid Affine / Sparse-Polygon decision
 - This Graphics White Paper v2.1

- MUST:** This document is subordinate to Architecture 1.5.1, Engine 0.2, and F65-GAD-001.
- MUST:** Where this document conflicts with a parent, the parent wins and this document must be corrected.
- MUST:** Exact resolution, viewport layout, face/vertex/span ceilings, LOD distances, RRB production selection, palette, cycle budgets, occlusion envelope resolution, registration-record size, and maximum displayed-world age remain **R0-GATED**. Candidate values are measured during R0-D/E, hardware-confirmed in R0-F, and frozen only in the measured-limits revision.
- MUST:** Graphics White Paper Rev 1.0 is supporting technical reference / feasibility material only. Its internal “Project Baseline” label was never issued through the F-65 authority chain.

0.1 R0 scheduling constraint

R0-A and R0-B are completed evidence milestones. R0-C implementation was substantially complete when Graphics v2.1 was issued. This document does **not** retroactively alter their implementation or acceptance scope.

Graphics requirements introduced or materially expanded by v2.1 (including TerrainOcclusionEnvelope, displayed-world-age measurement, atomic DisplayedWorldGeneration / view coherence, LOD hysteresis, registration-anchor priority policy as a measured behavior, terrain visual/query registration proof, positive-G desaturation presentation transform, and the occlusion/coherence torture scene) are integrated through **R0-D and R0-E** and confirmed by **R0-F**.

Existing R0-A/B/C evidence remains authoritative for the scope actually tested. Do not reopen, retrofit, or rewrite R0-A, R0-B, or R0-C acceptance for Graphics v2.1.

Gate	Graphics v2.1 disposition
R0-A	NO CHANGE. CLOSED.
R0-B	NO CHANGE. CLOSED. Existing graphics/display evidence remains valid for exactly what it tested.
R0-C	NO CHANGE. FINISH CURRENT WORK. No graphics retrofit.
R0-D	Add measurement hooks and protected-load fixtures needed to account for the new graphics behavior. Do not turn D into a second renderer-development phase.
R0-E	Primary implementation/integration gate for the new Graphics v2.1 requirements. Implement and exercise them in the combined Xemu harness.
R0-F	Confirm the same identified R0-E configuration on physical MEGA65. Hardware-specific corrections may produce a new identified build and required E→F revalidation; a new renderer architecture must not first appear here.

This allocation matches the program definition: R0-D is the protected-workload/calibration milestone; R0-E combines renderer, snapshot, memory, input, audio, DMA/IRQ, storage, faults and shedding in Xemu; R0-F is physical-hardware confirmation of an identified R0-E configuration.

1. Design goals and non-goals

1.1 Primary goals

- Horizon stability under continuous attitude change, driven exclusively from PresentationSnapshot / world-registration state.
- Consistent, measurable frame-time behavior and bounded displayed-world age rather than headline FPS claims.
- Convincing visual range via layered techniques (analytic sky, silhouettes, affine surface, sparse geometry, impostors) with a defined cross-layer occlusion contract.
- Functional, readable cockpit instruments and HUD that remain deadline-protected and spatially coherent with the displayed world.
- Support for internal (cockpit) and external (chase) camera modes without forcing unfinished world buffers to display, and without mismatched view composition.
- Deterministic residency and graceful LOD fallback; no frame may stall waiting for optional visual data.
- Aspect-faithful aircraft representation at combat ranges, with deterministic LOD hysteresis.
- Spatial registration between the displayed outside world and world-registered HUD symbology, even when the world buffer is intentionally stale.
- Visual terrain and carrier geometry that remain registered with authoritative query/contact geometry within approved tolerances.

1.2 Explicit non-goals

- Full 6DOF free-look dense polygonal scenes.
- Perspective-correct texturing of arbitrary meshes as a production requirement.

- Photorealistic lighting or dynamic shadows.
- Continuous predictive Attic streaming as a dependency for frame completion.
- Billboard-only aircraft as the primary combat representation.
- A second independent renderer architecture parallel to Engine 0.2.
- Treating prior R0-B candidate measurements as automatic production selection of RRB, FCM/NCM/SEAM, or any specific VIC-IV mode.
- Frame rates above the measured protected-load capability of the platform.
- A Z-buffer.
- Automatic promotion of diagnostic/proof shedding tiers (including wireframe) into player-visible production behavior.

1.3 Mandatory F-65 presentation contracts

Architecture 1.5.1 already requires the following player-facing presentation contracts. Graphics implementation must satisfy them:

- **Cockpit-primary presentation** with a bubble-canopy visual character and a low-cost chase view.
- **Green monochrome HUD** as the primary flight-data source.
- **Grayscale aircraft-status display** (left cockpit display character as defined by architecture/gameplay).
- **Positive-G desaturation** beginning near +6 G and reaching grayscale near +7 G.

G-desaturation rules:

- Desaturation is a presentation transform driven by the appropriate PresentationSnapshot (or status) state.
- It does not alter simulation, sensor, weapon, or checksum behavior.
- It must preserve critical HUD and warning readability.
- HUD green / critical symbology behavior follows the approved presentation design and must not be accidentally washed out by a whole-screen palette effect.

Exact implementation (palette transform, alternate palette banks, selective role protection) remains **R0-GATED**.

2. Presentation ownership and timing model

2.1 Authority boundary

- Simulation runs at exactly 100 Hz while the sortie is in ACTIVE_SORTIE.
- The renderer never integrates aircraft attitude or position dynamics.
- Presentation consumes only complete published PresentationSnapshot records.
- A world-buffer attempt binds one immutable source_snapshot_tick, one view, one presentation tier, one destination store, and one resumable work cursor.
- A single world buffer cannot mix entity or camera state from different snapshots.

2.2 Independent clocks and services

Service	Cadence / ownership
Simulation	Fixed 100 Hz
Protected presentation (HUD compositor, cockpit compositor, critical warnings, display publication/housekeeping)	Every display service
Radar symbology / systems pages / lower-priority instruments	Approved scheduled cadences (may remain visually present without recomputation every frame)
World-frame construction	Remaining time only; incremental and resumable

Service	Cadence / ownership
Audio	Independent service contract

2.3 Overrun and shedding

When execution falls behind:

1. World rendering yields.
2. Decorative presentation is suppressed according to the fixed shedding ladder.
3. The last completed world buffer remains displayed.
4. Simulation, input, audio, HUD, and critical warnings continue.
5. Tick debt is processed without reordering.

Shedding changes only presentation work. It never changes sensor visibility, collision geometry, AI perception, missile state, mission state, or checksums.

Shedding ladder vs production tiers:

The fixed shedding ladder (Engine 0.2) defines technically validated fallback candidates, including reduced-filled and wireframe proof tiers. R0-E/F evidence and product acceptance determine which tiers are permitted in shipping gameplay. A diagnostic or proof tier is **not** automatically a player-visible production tier. Sustained need to reach wireframe under an approved normal load is more likely evidence that the selected graphics configuration failed than permission to ship wireframe as normal degradation.

2.4 World-frame construction job

A world-frame construction job is incremental and resumable and may span multiple display services. Only a completed world frame is published for display. Partial work is never shown.

RenderWorkItem carries the stage, snapshot tick, view/tier identity, resource handles, object/face/span cursor, destination bounds, pending DMA identity, and diagnostic counters needed to resume without repeating or skipping work. Yielding occurs at tile, object, face, span-batch, or DMA-batch boundaries.

2.5 Displayed-world age and latency

Architecture treats sustained world cadence below **20 Hz** as a failure floor, not permission to alter simulation. Graphics must also measure how old the displayed picture is.

A renderer that completes frames at 20–25 Hz but regularly displays world state 100–150 ms behind the simulation is materially different from a fresh 25 Hz image.

R0-D must instrument and R0-E must record under combined load at least:

Metric	Meaning
world_frame_construction_time	Time from bind to complete for a world-frame job
displayed_world_age_ticks	current_simulation_tick – source_snapshot_tick of the displayed buffer
displayed_world_age_ms	Age in milliseconds
source_snapshot_tick	Bound snapshot of the displayed (or completing) buffer
current_simulation_tick	Live simulation tick at sample time
view_change_to_first_complete_frame_latency	Latency from view-change request to first complete frame for that view
p50 / p95 / worst	For the above under protected load

Maximum permitted displayed-world age is **R0-GATED**: candidate values measured in R0-D/E, hardware-confirmed in R0-F, frozen only in the measured-limits revision. Prefer consistency of age and frame time over peak FPS.

2.6 Atomic display generation and view coherence

A completed world store and its `worldRegistrationRecord` form one **indivisible display-generation pair**. Neither may become display-eligible without the other.

Each display generation carries common identity fields:

- `world_generation_id`
- `source_snapshot_tick`
- `view_id`
- `tier_id`

View-change coherence:

The currently displayed view and composition profile remain active until a complete frame for the requested view is ready. Then the new world buffer + `worldRegistrationRecord` + view composition profile swap together as one generation.

The protected compositor must not switch to left-glance canopy framing (or chase overlays) while still displaying an old forward world buffer, and must not put chase-view overlays over an old cockpit world buffer.

Status / warning / non-registered symbology may continue to update independently from the newest permitted snapshot.

Conceptually:

```

DisplayedWorldGeneration
  world buffer
  WorldRegistrationRecord
  view identity
  view composition profile

```

3. Hybrid layer architecture

The world is composed of ordered layers. Affine is a rendering primitive, not the world model.

Layer	Technique	Notes
Sky / atmosphere	Analytic fill, gradients, haze	Low cost; filled once per completed world frame after shared horizon state is known
Distant terrain / horizon	Ridge / silhouette strips, very-low-detail impostors or parallax	Visual range; contributes to <code>TerrainOcclusionEnvelope</code>
Ocean / broad flat terrain	Affine or scanline ground plane (DMA-accelerated where practical)	Primary high-range surface; consumes shared horizon/camera state
Gentle terrain relief	Bounded height modulation or selective silhouette features	Only where proven useful; not a full height-field mesh
Carrier deck + island	Sparse projected geometry	Core operational requirement; subject to registration invariant
Airfields	Affine/scanline where compatible; combat-critical markings, thresholds, lights, structures via registered overlay or sparse geometry	Best surface technique measured in R0-E
Ships / close structures	Sparse projected geometry	Same path as carrier
Aircraft	Point/glint → billboard → directional impostor → low-poly	Aspect fidelity required at combat ranges; LOD hysteresis required
Missiles / effects (visual)	Point, sprite, streak, billboard	Non-authoritative representation only
Protected presentation	VIC-IV composition; RRB retains its R0-B candidate evidence and remains a leading production candidate pending downstream D/E/F confirmation and measured-limits selection	Higher priority than world geometry

Exact screen-area or terrain-area balance between affine and sparse-geometry paths is measured in R0-E and is not frozen here.

3.1 Cross-layer occlusion contract

Depth buckets solve sparse-polygon versus sparse-polygon ordering. They do **not** by themselves solve sparse objects versus an affine surface or silhouette terrain.

Without an additional contract, a fighter can draw through a mountain ridge, a missile can pass incorrectly in front of terrain, or a ship hull can fail to intersect the water plane convincingly.

Presentation-only TerrainOcclusionEnvelope:

Distant terrain / ridge rendering (and any other layer designated as occlusion-generating) produces a bounded screen-space occlusion envelope for the current world-frame attempt. Typical content (exact resolution R0-gated):

- Horizon / terrain Y by screen column (or coarse column groups)
- Optional coarse distance / depth classification per column
- Validity flag and source_snapshot_tick / world_generation_id

Aircraft, impostors, missiles, and other sparse objects that must not draw through foreground terrain reject or clip presentation fragments that are visually behind the envelope.

- This envelope is **presentation-only**.
- Authoritative radar LOS, collision, terrain impact, ground clearance, and contact continue to use the world-query system, never rendered LOD or the occlusion envelope.
- Exact column resolution, generation cost, and clip/reject policy are **R0-GATED**.

Typical cases the contract must cover:

- Enemy aircraft flying behind a mountain ridge
- Missile passing behind terrain
- Carrier geometry intersecting the ocean surface
- Low aircraft passing behind an airport structure
- Ship hull intersecting the water plane

4. Affine / scanline surface path

4.1 Scope

The affine (or equivalent scanline) path is used for:

- Ocean and large flat expanses
- Compatible runway / taxiway base surfaces
- Distant terrain continuity where silhouette layers are insufficient

It is **not** required to represent steep terrain, true 3D mountains, carrier superstructure, ships, or close air-combat geometry.

Affine surface rendering **consumes** the shared camera / horizon projection state. It does not own horizon management.

4.2 Shared horizon / projection state

The horizon is shared view/projection state consumed by:

- Sky fill
- Affine ground / ocean
- Sparse geometry (as needed)
- HUD registration / artificial horizon
- TerrainOcclusionEnvelope generation

Its screen-space Y position and tilt are derived from the camera orientation published in the world-registration record of the displayed buffer (or the active construction snapshot while building). Small errors are immediately noticeable; the attitude-to-horizon transform must be high-precision fixed-point and free of visible jitter.

Ground-plane affine parameters and the sky fill region are both derived from this shared state.

4.3 Implementation notes (candidates)

- Per-row start U/V and dU/dX, dV/dX computed from camera and horizon.
- DMA fractional addressing preferred for horizontal spans where measurement shows advantage.
- Tight 45GS02 loops with Q-register operations remain valid candidates at 320-pixel widths.
- Distant rows may use coarser detail or haze/fog color lerp.
- Exact affine parameter formats, table sizes, and DMA vs CPU split remain R0-gated.

4.4 Integration with the sparse pipeline

The affine surface is composed early in the world-frame job (after sky/background bands) and does not replace the sparse polygon stages for objects that require projected geometry. Affine and sparse geometry share the same destination store, the same `source_snapshot_tick` binding, and the same `world_generation_id`.

5. Sparse polygon pipeline

This is the Engine 0.2 software 3D pipeline, confirmed as the sparse-geometry implementation basis. Graphics v2.1 does not create a parallel renderer.

5.1 Ordered stages

1. Acquire newest complete READY PresentationSnapshot permitted by the world-frame scheduler; bind destination store, `source_snapshot_tick`, `view_id`, and `tier_id`.
2. Clear or compose authored sky / ocean / background bands for the selected view and lighting preset.
3. Establish shared camera / horizon projection state for this generation.
4. Rebase extracted presentation positions to signed camera-relative coordinates.
5. Gather potentially visible terrain tiles and object presentation records using compiled conservative bounds.
6. Select permitted mesh or impostor LOD from the current fixed shedding tier, applying deterministic LOD hysteresis (§6.4).
7. Transform model vertices into camera space with widened, saturating fixed-point intermediates.
8. Reject objects outside the conservative frustum; backface-cull eligible faces; clip surviving geometry against near plane and required side/top/bottom planes.
9. Apply TerrainOcclusionEnvelope reject/clip for objects that must not draw through foreground terrain (§3.1).
10. Perspective-project clipped vertices into bounded screen coordinates; record every saturation or clip-capacity rejection.
11. Convert faces and impostors into bounded render primitives (palette/material role, depth key, edge/span metadata, source identity).
12. Place opaque primitives into coarse far-to-near depth buckets. Apply compiled subobject order or bounded local sort where carrier/runway geometry requires it.
13. Traverse terrain and opaque-object buckets far to near; generate edges with one frozen inclusion rule; emit clipped spans or validated DMA batches.
14. Draw ordered transparent-index impostors and presentation effects without changing the authoritative depth/query model.
15. Generate / finalize TerrainOcclusionEnvelope contribution for this generation if required by later consumers.
16. Mark the world store complete only after every required CPU span and immutable DMA batch completes.
17. Publish WorldRegistrationRecord atomically with the completed buffer (§7.3).
18. Offer the complete display-generation pair for swap at a display boundary under the atomic view-coherence rule (§2.6); compose protected cockpit/HUD layers according to §7; release records only at legal ownership transitions.

5.2 Explicit constraints

- No hardware or software Z-buffer.

- No requirement for perspective-correct texture mapping.
- Every pool (vertices, faces, spans, objects, anchors, occlusion columns) has fixed capacity and defined overflow behavior.
- Overflow is deterministic: excess geometry is dropped or reduced by frozen policy; it does not corrupt the frame or steal protected work.
- Clip-list, vertex, face, bucket, edge, span, or DMA-list exhaustion aborts the incomplete buffer, increments diagnostics, and steps down the fixed presentation ladder.
- Edge generation uses a single top/left-versus-bottom/right inclusion convention so adjacent faces neither double-fill nor open presentation cracks.
- Intersecting local geometry that cannot tolerate coarse painter ordering (especially runway/carrier presentation) uses authored subobject ordering or bounded local sorting.

5.3 Deterministic ordering

Ordering keys eventually include stable tie-breakers:

```
depth bucket
  → object / render priority
  → stable object handle
  → face / index order
```

Exact key definitions remain R0-gated. Ties must resolve deterministically and independently of incidental traversal order.

5.4 DMA discipline

- World-render DMA is subordinate to protected display/system DMA.
- World DMA submissions must be bounded into latency-safe batches.
- A DMA batch is range-validated, immutable until completion, and submitted only through core-owned DMAService.
- Admission includes measured blocking time so the renderer cannot submit work that violates protected deadlines.
- Renderer yielding includes DMA-batch boundaries.

6. Aircraft, objects, and LOD ladder

6.1 Aircraft representation ladder

Range class (conceptual)	Representation
Extreme distance (1–3 px)	Point / glint
Far	Billboard / silhouette
Medium	Directional impostor (multi-angle sheet)
Near	Low-poly mesh
Very near / external close-up	Low-poly + limited detail

Billboard-only is rejected as the primary combat strategy. Aspect fidelity at combat ranges requires actual geometry (or high-quality directional impostors) once the object occupies enough screen space for aspect to matter.

6.2 Other dynamic objects

- Ships and carrier: sparse projected geometry with registration anchors.
- Missiles / effects: point, sprite, streak, billboard as appropriate. Visual representation only; presence, omission, LOD, or update rate cannot change authoritative missile, aircraft, weapon, sensor, collision, or mission state.
- Presentation effects (flame, smoke, contrail, explosion flash, etc.): presentation-only unless a later numbered decision grants physical influence.

6.3 LOD rules

- Every renderable aircraft, ship, significant world object, and effect declares mesh LODs, impostor forms, palette roles, bounds, and switching metadata at compile time.
- Impostors and cheap LODs consume explicit asset bytes, visible-list entries, transform work, DMA work, and draw clocks in the same ledgers as filled geometry.
- No report may call an impostor “free.”
- LOD choice is presentation-only and cannot change contact identity, collision bounds, sensor return, damage, or mission behavior.
- Exact mesh ceilings, impostor dimensions, angle bins, transition distances, and maximum visible-object counts are R0-gated.

6.4 Deterministic LOD hysteresis

An aircraft hovering around one screen-size threshold must not alternate impostor ↔ mesh every frame. A directional impostor must not flip between neighboring angle bins as aspect oscillates slightly.

Required behavior:

- Separate enter and exit thresholds for each LOD transition.
- Stable angle-bin hysteresis for directional impostors.
- One chosen representation for the entire world-frame attempt (no mid-frame switching merely because the renderer resumed later).
- Deterministic selection from the bound snapshot and stable object identity.

Exact thresholds and bin widths remain **R0-GATED**.

7. Protected presentation, HUD, and views

7.1 Protected versus scheduled

Protected every display service:

- HUD compositor
- Cockpit compositor
- Critical warnings
- Display publication / housekeeping

Scheduled independently:

- Radar symbology and data refresh
- Systems pages
- Lower-priority instruments

7.2 World-registered versus status symbology

World-registered (must stay aligned with the displayed outside world):

- Pitch ladder / attitude reference
- Flight-path marker
- Target designators
- Weapon pippers / steering cues
- Other geometry registered against the outside scene

Status / non-registered (no spatial registration requirement):

- Airspeed, altitude, Mach, fuel
- Warnings and cautions
- Weapon counts

- Mode / status text

7.3 WorldRegistrationRecord (atomic with world buffer)

Every completed world buffer publishes a bounded immutable **WorldRegistrationRecord** derived from the same source snapshot used to construct that frame.

A completed world store and its WorldRegistrationRecord form one **indivisible display-generation pair**. Neither may become display-eligible without the other. They share `world_generation_id`, `source_snapshot_tick`, `view_id`, and `tier_id`.

The record contains the minimum camera / projection / horizon state and bounded spatial anchors required to compose world-registered symbology for as long as that world buffer remains eligible for display. It does **not** require the original PresentationSnapshot buffer to remain acquired after its legal ownership lifetime.

Candidate contents (exact fields R0-gated):

- `world_generation_id`
- `source_snapshot_tick`
- `view_id / tier_id`
- camera position / rebase origin
- camera orientation or display transform
- projection parameters
- horizon / reference parameters
- bounded projected anchors for applicable target designators, steering cues, weapon/pipper references, and other world-registered entities

Where a spatial cue references a rendered world object, the completed frame retains an appropriate bounded registration anchor or projected reference derived from the same world-frame state. The protected compositor shall not combine the displayed position of an object from one snapshot with a newer authoritative or presentation position in a way that visibly breaks registration.

7.4 Registration-anchor overflow priority

Registration anchors are a fixed-capacity pool. When the pool fills, drop order must be deterministic and priority-aware:

1. **Highest:** selected target / weapon solution / essential flight cue
2. Threat / tactical cues
3. Secondary tracks
4. **Lowest:** decorative / nonessential world labels

Exact capacities and priority encodings remain **R0-GATED**. It is unacceptable to drop the selected target designator because decorative formation markers were inserted first.

7.5 Supported views

- Forward cockpit
- Fixed left glance, fixed right glance, fixed aft-quarter glances
- Low-cost chase view

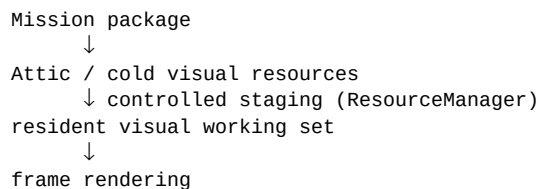
Glance views use authored camera transforms and cockpit framing. A freely rotating camera is not required. View changes cannot alter simulation, AI, radar, or checksums and cannot force an unfinished buffer to display. View composition swaps only with a complete matching display generation (§2.6).

7.6 Composition candidate

VIC-IV character modes and Raster Rewrite Buffer (RRB) were preferred R0-B composition candidates and retain that evidence value. Production selection remains **R0-GATED** and is not frozen by this white paper; confirmation proceeds through downstream D/E/F and the measured-limits revision.

8. Residency, fallback, and streaming

8.1 Baseline residency model



A valid frame must be producible from the currently admitted resident visual working set without blocking on an unavailable Attic resource. The full mission visual package may live in Attic; the resources required for the current frame must be deterministic and available.

8.2 Mandatory fallback

Every streamable visual asset class that may be absent from the current resident set must define a resident fallback representation. Examples:

- detailed terrain → coarse terrain
- low-poly aircraft → directional impostor or billboard
- directional impostor → billboard
- billboard → point / glint
- detailed structure → coarse hull / silhouette

A missing optional high-detail resource causes deterministic LOD fallback, not a stalled frame.

8.3 Streaming policy

Continuous predictive Attic streaming is an optional enhancement after the resident path is proven. It must never be required to complete a frame or to keep the horizon stable. Tactical play performs no disk reads.

8.4 Carrier visual / contact registration

Carrier visual geometry and carrier-contact geometry share the same canonical carrier transform, deck reference frame, and authored reference landmarks. Rendering geometry need not equal collision geometry polygon-for-polygon, but deck edges, landing area, touchdown references, catapult locations, arresting geometry, and island placement must remain spatially registered within an approved tolerance.

8.5 Terrain visual / query registration

Presentation terrain resources, coastlines, runways, exceptional terrain features, and their authoritative query counterparts share the same canonical world/tile transforms and authored registration references. Presentation LOD may simplify or disappear under shedding, but may not move terrain/reference geometry outside approved visual/query tolerances.

Authoritative terrain height, collision, line-of-sight, and contact **never** derive from rendered LOD or from the TerrainOcclusionEnvelope. Sensor LOS and contact continue to use the world-query system as required by Engine 0.2.

This invariant is especially important for low-altitude flight, ridge crossings, and runway elevation.

9. Memory and workspace

Pre-R0 ownership (from Architecture 1.5.1 / Engine 0.2) relevant to graphics:

Physical range	Size	Binding use
\$020000–\$02FFFF	64 KB	Display store A
\$030000–\$03FFFF	64 KB	Display store B

Physical range	Size	Binding use
\$040000–\$047FFF	32 KB	Cockpit base, HUD/MFD glyphs, hot sprites/impostors, palettes
\$048000–\$04FFFF	32 KB	Transformed vertices, faces/LOD/depth buckets, edges/spans, visible lists, renderer DMA staging, occlusion workspace (as measured)
\$050000–\$057FFF	32 KB	Tile/resource decode staging, audio sample cache, resident DMA lists

Exact sharing between cockpit assets, hot impostors, renderer records, tile staging, occlusion envelope storage, and audio sample cache freezes after simultaneous display/audio/DMA measurement. PresentationSnapshot maximum bytes, buffer count, payload location, WorldRegistrationRecord size/location, and occlusion-envelope resolution remain R0-gated.

Renderer code shares the resident code ceiling with other modules; the pre-R0 renderer/cockpit/HUD/radar/systems-display service ceiling is 6 KB of the sortie code envelope (subject to measured rebalance).

10. Asset pipeline notes

- Blender / glTF 2.0 is the primary source path for aircraft, ships, cockpit geometry, significant features, and authored mesh LODs.
- Heightfield sources use documented lossless formats; 2D art uses lossless PNG plus palette-role metadata.
- Host compilers quantize coordinates, generate/validate bounds, validate or generate permitted LOD/impostor forms, enforce palette roles, encode target resources, and report decoded/staged sizes.
- Target code never parses authoring formats.
- Every asset declares stable ID, owner, provenance, conversion recipe, palette role, converted-size limit, residency/preload class, and fallback/proxy.
- Converters reject out-of-envelope assets rather than truncate them.

Offline conversion to indexed color and binary blobs consumable by DMA remains the practical path for terrain tiles and character assets. No runtime parsing of complex formats.

11. Implementation language

Consistent with Architecture 1.5.1 and Engine 0.2:

1. Define the behavioral, numeric, ownership, memory, timing, and fault contract.
2. Implement or update independent Java oracle and golden vectors where applicable.
3. Implement the target routine in C unless a platform/hardware invariant justifies low-level code immediately.
4. Measure target-code bytes, data/constant/runtime bytes, stack high-water, cycles, DMA interaction, and hardware behavior.
5. Retain C when it fits every approved budget and contract.
6. Identify a measured offender before replacing or supplementing with handwritten 45GS02 assembly.
7. Require the replacement to satisfy the same public contract and validation suite.

Likely ASM candidates after measurement:

- Span generation / fill
- Clipping inner loops
- Perspective divide / table lookup
- Affine scanline setup
- DMA job creation
- Matrix / vector kernels

- Occlusion-envelope generation / test
- Screen clear / fill

Object traversal, LOD selection (including hysteresis), render queues, clipping orchestration, camera management, and frame scheduling begin in C. No fixed C/assembly percentage is required.

12. Performance tiers and measurement

Headline FPS tables are not frozen. Use tiers:

Tier	Intent
Target	Normal world presentation under typical load
Acceptable	Under combat load
Degraded	Temporary stress
Failure floor	Sustained world cadence below 20 Hz under the protected R0 workload is failure (Architecture 1.5.1), not permission to alter simulation

Prefer frame-time consistency and bounded displayed-world age over peak FPS. A stable lower rate with controlled age is preferable to alternating high/low rates or a “fast” but stale picture.

Legacy two-display-frame planning figures remain planning allocations only. Production ceilings freeze through R0 measurement and the measured-limits revision.

Fixed presentation shedding ladder (Engine 0.2) — **technical candidates only**:

1. Drop optional presentation effects.
2. Select cheaper distant-object impostors.
3. Reduce distant terrain detail.
4. Remove optional shading and decorative faces.
5. Fall from filled to validated reduced-filled and then validated wireframe **proof** tiers.
6. Retain the previous completed world buffer and reduce world-frame cadence.

Shipping-permitted tiers are a product/R0 acceptance decision. Wireframe is not automatically a player-visible production degradation tier (§2.3).

13. Hardware capabilities relevant to the hybrid model

Useful VIC-IV and platform features (candidates, not frozen requirements):

- Full Color Mode (FCM) and Nibble Color Mode (NCM) for ground and sky textures.
- Super Extended Attribute Mode (SEAM) for large character sets and per-character attributes.
- Raster Rewrite Buffer (RRB) with GOTOX — R0-B candidate with retained evidence; production use subject to downstream D/E/F confirmation and measured-limits selection.
- Multiple 256-color palette banks (including roles protected under G-desaturation).
- Hardware scaling and flexible border positioning for viewports.
- DMA with fractional address stepping for affine spans and tile staging.
- 45GS02 32-bit addressing, Q-register, and MAP banking for matrix work and large working sets.

Exact mode selection, palette layout, and DMA vs CPU split are R0-gated.

14. Testing and validation criteria

- Primary development in emulator with frequent real-hardware validation.
- Instrument frames with raster-position timing; log worst-case frame times, high-water marks, and displayed-world age.
- Visual tests: pure pitch, pure roll, coordinated turns, high-rate attitude changes — expose horizon jitter or affine errors.
- Registration tests: intentionally stale world buffers with active world-registered symbology; verify pitch ladder, FPM, and target boxes remain aligned with the displayed image while status updates continue.
- Distance tests: fly away from a known airport/carrier and verify smooth LOD transitions (with hysteresis) and haze behavior.
- Carrier approach tests: visual deck edges, landing area, and island placement remain registered with contact geometry within approved tolerance.
- Terrain registration tests: ridge height, runway elevation, and coastlines remain within visual/query tolerances; authoritative queries unchanged by rendered LOD.
- Stress: maximum simultaneous traffic + active runway/carrier overlays + full instrument set + presentation effects under the combined-load harness.
- Incomplete buffers never display; camera/view/tier changes never alter checksums; view composition never mismatches the displayed world view.
- Forced presentation lag proves FREE/PUBLISHING/READY/READING ownership, no torn record, no simulation stall, and unchanged checksums.

14.1 Occlusion / coherence torture scene (R0-E primary; R0-F hardware confirmation)

A single combined scene should exercise:

- Aircraft approaching / descending behind a ridge
- Aircraft crossing ridge silhouette
- Aircraft crossing ocean horizon
- Ship partially intersecting waterline
- Carrier approach with island / deck geometry
- Selected target designator active
- Intentionally stale displayed world buffer
- LOD transitions during the crossing
- View change during an unfinished frame
- G-desaturation active (if fixture permits)

Verify:

1. No target draws through foreground terrain incorrectly.
2. Terrain / collision registration stays within tolerance.
3. Target box stays registered to the displayed image.
4. No LOD flicker (hysteresis holds).
5. No partial frame displays.
6. View composition never mismatches world view.
7. Current warnings / status keep updating.
8. World-buffer age remains within measured limits.
9. Simulation checksum is unchanged.
10. Anchor priority preserves selected target under pool pressure (if exercised).

15. Limitations, risks, and mitigations

Risk	Mitigation
No Z-buffer; approximate depth ordering	Deterministic depth buckets + stable tie-breakers + authored subobject order for carrier/runway + TerrainOcclusionEnvelope for cross-layer cases
Affine cannot represent steep terrain or true 3D mountains	Silhouette layers + selective sparse features; affine scoped to ocean/flat/compatible surfaces
Chip RAM working-set limits high-detail terrain and unique billboards	Mission-local resident working sets + mandatory fallback LOD
Attic streaming latency	Streaming optional; resident path must complete valid frames without it
Close external views of detailed aircraft meshes	Prefer billboards/impostors except where measurement admits low-poly near LOD
Snapshot-age disagreement between world and HUD	WorldRegistrationRecord lifetime + atomic display-generation pair + world-registered vs status split
View change vs unfinished world	Atomic DisplayedWorldGeneration swap; old view remains until new complete generation
Stale but “complete” world frames	Explicit displayed-world age metrics; R0-gated maximum age
LOD flicker at thresholds	Deterministic enter/exit hysteresis and per-frame-attempt lock
Anchor pool overflow drops critical cues	Priority-ordered drop policy (selected target highest)
Large world DMA monopolizing the engine	World DMA subordinate, latency-safe batches, yield at DMA-batch boundaries
Over-ambitious geometry budgets	Geometry admitted only by measurement; fixed capacities and deterministic overflow
Wireframe appears in normal play	Shedding ladder is technical; production tiers are product/R0 acceptance

Treat the hybrid affine + sparse + impostor + protected composition design as the baseline. Add complexity only after the core horizon-stable, registration-coherent, occlusion-correct, resumable pipeline meets its frame-time, age, and visual stability targets under the combined-load harness.

16. Downstream R0-D/E/F graphics adoption and proof requirements

R0-A and R0-B are closed. R0-C is not reopened by this document. New or expanded Graphics v2.1 requirements are admitted only through the allocation in §0.1.

16.1 Existing R0-B evidence — no retrofit

R0-B graphics/display evidence remains authoritative for the scope it actually tested (display candidates, cockpit composition experiments, double buffering, palette behavior, representative renderer candidates, input latency/edge, and related measurements as recorded in the R0-B identity). This white paper does not require re-running or expanding that scope.

Capabilities already exercised under R0-B (horizon-stable affine candidates, sparse polygon candidates, protected composition candidates, resumable complete-buffer publication, basic registration ideas, resident fallback concepts) retain their evidence value. They are not invalidated; they are also not expanded retroactively.

16.2 R0-D instrumentation and calibration additions

Keep Delta **narrow**. Purpose: ensure the historical protected workload can account for the cost of Graphics v2.1 behaviors before they enter the combined harness. Delta is **not** “re-run R0-B with the new renderer.”

Add measurement hooks and protected-load fixtures as needed for:

- world_generation_id
- source_snapshot_tick
- displayed-world-age counters (ticks and ms)
- world-frame construction-time counters
- view-change-to-first-complete-frame latency counter
- occlusion workload / high-water counters

- registration-anchor high-water / overflow counters
- LOD transition counters
- DMA latency / blocking contribution attributable to world-render batches
- protected compositor cost with G-desaturation candidate active

R0-D establishes the protected-load baseline accounting for these costs. It does not implement the full Graphics v2.1 integration scene set.

16.3 R0-E graphics integration requirements (primary gate)

Echo is the **primary implementation and integration gate** for Graphics v2.1 requirements introduced or materially expanded after R0-B. Implement and exercise in the combined Xemu harness:

1. **TerrainOcclusionEnvelope** and sparse-object reject/clip behavior (presentation-only; authoritative LOS/collision remain with the world-query system).
2. **Atomic DisplayedWorldGeneration** — world buffer + WorldRegistrationRecord + view ID + composition profile + common generation identity; atomic swap semantics.
3. **Displayed-world-age** measurement and maximum-age candidate under load (distinguish cadence from stale-image latency).
4. **Deterministic LOD hysteresis** (enter/exit thresholds, angle-bin hysteresis, one representation per world-frame attempt).
5. **Registration-anchor priority / overflow** behavior under pool pressure.
6. **Terrain visual/query registration** proof (canonical transforms; LOD may simplify, not move geometry; queries never from rendered LOD).
7. **Positive-G desaturation** presentation transform with protected HUD/warning readability (architecture visual contract; graphics implementation measured here).
8. **Atomic view changes** (no mismatched canopy/chase composition over an old world buffer).
9. **Occlusion / coherence torture scene** (§14.1).
10. **Combined-load cost** of all of the above with snapshot, memory, input, audio, DMA/IRQ, storage, faults, and shedding.

No R0-E result freezes a production TARGET, TBD, or R0-GATED value without the measured-limits revision.

16.4 R0-F physical-hardware confirmation

Foxtrot uses the **exact pinned R0-E build/configuration** and answers:

- Does TerrainOcclusionEnvelope behave correctly on actual VIC-IV timing?
- Are DMA batches latency-safe under real hardware blocking?
- Does displayed-world age remain bounded?
- Does view switching remain atomic?
- Are HUD/world anchors stable on hardware?
- Does LOD hysteresis behave correctly?
- Does G-desaturation preserve HUD/warning readability?
- Does terrain/carrier registration visually hold?
- Are p95 / worst / high-waters / reserve still acceptable?

R0-F is **not** where new renderer architecture first appears. If physical hardware exposes a defect that forces a correction, that correction produces a new identified build and triggers the required E→F revalidation under existing program rules.

16.5 Closure of R0-GATED graphics values

For Graphics v2.1—introduced R0-GATED items (occlusion-envelope resolution, registration-record size/location, maximum displayed-world age, LOD hysteresis thresholds, anchor capacities/priorities, production shedding tiers permitted in shipping, and related limits):

Step	Gate
Candidate values measured	R0-D / R0-E
Hardware confirmed	R0-F
Frozen	Measured-limits revision only

17. Implementation roadmap (program timeline)

Completed / immutable for this revision

R0-A
R0-B

In completion – do not disturb for Graphics v2.1

R0-C
no Graphics v2.1 retrofit

Next

R0-D
instrumentation + protected-load accounting for v2.1 costs
(narrow; not a second renderer-development phase)

Then

R0-E
Graphics v2.1 integration in Xemu combined harness
TerrainOcclusionEnvelope
displayed-world age
atomic DisplayedWorldGeneration / view coherence
LOD hysteresis
registration-anchor priority
terrain visual/query registration
G-desaturation presentation
occlusion/coherence torture scene
combined-load cost of the above

Then

R0-F
exact corresponding physical-hardware confirmation
of the identified R0-E configuration

Then

measured-limits revision
freeze R0-GATED graphics values on evidence

Post-measured-limits product work (airport detail polish, chase/glance content, atmosphere presentation polish, measured C→ASM optimization of hot kernels) proceeds under normal Phase gates and does not reopen R0-A/B/C.

Every intermediate milestone must remain measurable and must never display an incomplete world buffer or a mismatched view composition.

18. Relationship summary

Document	Relationship
Architecture 1.5.1	Controlling parent — timing, ownership, snapshot, overrun, memory map, 20 Hz world failure floor, visual presentation contracts (HUD green, G-desaturation, canopy character)
Engine 0.2	Controlling engineering design — sparse pipeline stages, RenderWorkItem, shedding ladder, memory subledgers, terrain query vs presentation separation

Document	Relationship
F65-GAD-001	Controlling graphics architecture decision — hybrid layers, residency, registration, DMA priority, carrier invariant
Graphics White Paper Rev 1.0	Supporting reference only — VIC-IV research, RRB candidacy, DMA notes, horizon stability discussion retained for reuse
This document (v2.1)	Candidate detailed design under the above; occlusion, age, view coherence, hysteresis, visual contracts; new requirements scheduled through R0-D/E/F only (R0-A/B/C not retrofitted)

Appendix A. Key registers and platform notes (candidates)

Registers and features most relevant to the hybrid pipeline (consult official chipset documentation for full bit definitions):

Feature	Role
\$D02F (\$47/\$53)	Enable VIC-IV register set
\$D031	H640/V400, ATTR, display control
\$D054	CHR16 (SEAM), FCM enable bits
\$D058/\$D059	LINESTEP — logical row stride
\$D05E	CHRCOUNT — characters processed per row
\$D070	Palette bank select (including protected HUD / G-effect roles)
RRB / GOTOX	R0-B candidate with retained evidence; production use subject to downstream D/E/F confirmation and measured-limits selection
DMA list + fractional step	Affine ground row fills and tile staging
FNRASTER	Precise frame timing and interrupt points

Exact use remains subject to R0 measurement against the pinned MEGA65 Chipset Reference.

Appendix B. References

- F-65 Megawing Revision 1.5.1 — Architecture Invariants
- F-65 Engine Runtime and Toolchain Design Supplement Draft 0.2
- F65-GAD-001 — Hybrid Affine / Sparse-Polygon World Renderer
- Official MEGA65 Chipset Reference — VIC-IV and DMA chapters
- Dan Sanderson, MEGA65 Digest series (FCM, NCM, SEAM, Raster Rewrite Buffer)
- Classic Mode-7 / affine mapping literature (adapted to character + DMA constraints)
- MEGA65 Flight Simulation 3D Graphics White Paper Rev 1.0 (supporting reference)

End of Document — F-65 Megawing Graphics White Paper v2.1